As Fraunhofer diffraction is easy to observe and analyse, let us take it up for further discussions.

7.4.2. Diffraction in single slit

Let a parallel beam of light (plane wavefront) fall normally on a single slit AB of width a as shown in Figure 7.17. The diffracted beam falls on a screen kept at a distance D from the slit. The center of the slit is C. A straight line through C perpendicular to the plane of slit meets the center of the screen at O. Consider any point P on the screen. All the light reaching the point P from different points on the slit make an angle θ with the normal CO.

All the light waves coming from different points on the slit interfere at point P (and other points) on the screen to give the resultant intensities. The point P is in the geometrically shadowed region, up to which the central maximum is spread due to diffraction as shown Figure 7.17. We need to give the condition for the point P to be of various minima.

The basic idea is to divide the slit into even number of smaller parts. Then, add their contributions at P with the proper path difference to show that destructive interference takes place at that point to make it minimum. To explain maximum, the slit is divided into odd number of parts.

Condition for P to be first minimum

Let us divide the slit AB into two halves AC and CB. Now the width of each part is $a/2$. We have different points on the slit which are separated by the same width $a/2$ called as *corresponding points*. This is shown in Figure 7.18.

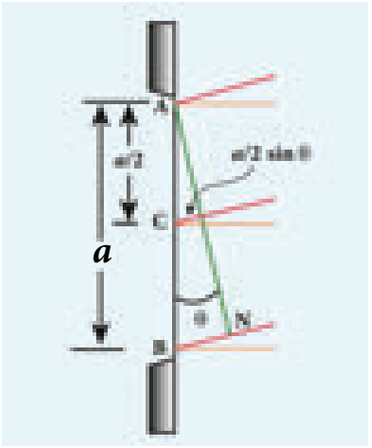


Figure 7.18 Corresponding points

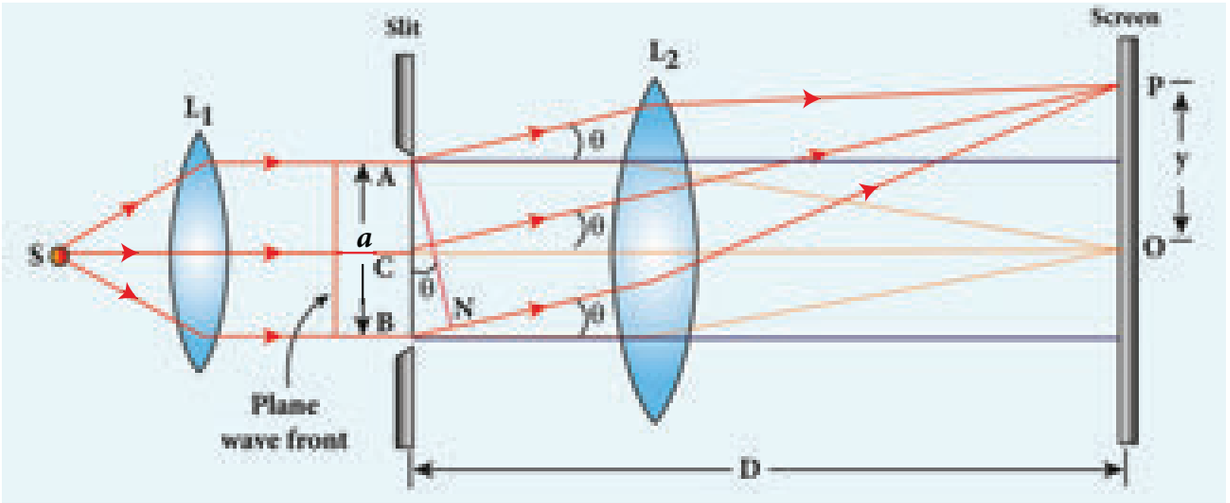


Figure 7.17 Diffraction at single slit

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